IRRIGATION WATER MANAGEMENT · NOTE WM04

Field Channels — Cross-Sections, Earth and Lined Channels, Design

Course topics: IDE5.5 L7, IDE8.17 L10, IDE-501 T8, IDE-503 T6

Choosing the cross-section

- Aim for the largest hydraulic radius the conditions allow. The most efficient theoretical section is the **semicircle** (least wetted perimeter, greatest hydraulic radius).
- Best rectangular section: depth = ½ × breadth.
- Semicircular and rectangular sections are practical only in brick or stone masonry and concrete-lined channels. Precast RCC elliptical sections (**canalets**) serve minor canals up to 1000 litres/sec.
- Small channels (≤ 60 litres/sec) are best rectangular: no side embankments, less land, fewer breaches from settlement, less weed growth, and cattle avoid crossing them.
- Large lined channels and earth channels are trapezoidal; the best hydraulic section is $b = 2\,d\,\tan(\theta/2)$ (b = bed width, d = depth of flow, θ = angle between side and horizontal). Real sections depart from it because of soil, topography and construction.
- Lay irrigation channels along the ridge line so water can be turned to either side.

Earth channels

Channels without any artificial lining. **Pros:** very low first cost, traditional, built and maintained by unskilled labour with no special materials. **Design concerns:** stable side slopes and non-erosive velocity — few natural materials tolerate more than 1.5 m/s.

Soil (Table 7.1)	Maximum permissible velocity (cm/s)
Loamy sand	45
Sandy loam	60
Loam, silt loam, silt	75
Clay loam	90
Gravelly soil and clay	100

- **Side slopes:** within the angle of repose — permanent channels not steeper than 1½ : 1; stiff clay up to ½ : 1; loose sand about 2 : 1 (horizontal : vertical).
- **Bed slope:** set mainly by the land slope — normal about 0.1%; silting starts below 0.05%. On steep land use drop structures or raise the bed, because steeper slopes mean higher velocity and erosion.
- **Alignment:** change direction smoothly and change section gradually.
- **Drawbacks:** heavy seepage; low velocity from rough surfaces; erosion; large cross-section; breaches from erosion and burrowing animals; moss and aquatic weeds; much maintenance labour.

Lined channels

- Lining overcomes these drawbacks, above all seepage in sandy soils and rodent-infested areas. It cuts seepage, protects against breaks, reduces weeds and moss, stops rodent damage and carries

more water for the same size.

- Lined channels sit on compacted beds with suitable slope; their commanding position lets them irrigate more land, and they need less maintenance labour. Whether lining pays depends on physical factors and costs; erosion is seldom a concern unless abrasive material is carried.
- **Materials:** concrete; stone or brick masonry; natural low-permeability clays; clay tiles; rubber, plastic and asphalt compounds. Good cement-concrete lining and single-layer brick or stone masonry in cement or lime mortar are virtually waterproof. Earth linings suit low budgets where suitable soil exists — a blanket of impervious clay over or within the bed and sides, or clay dispersed in the water to filter out with the seepage.

Concrete lining

Concrete is used more than any other material and, properly built, lasts long with little repair. Place it with wooden forms (Fig. 7.14) or apply it as plaster on the sides and bottom.

1. Compact the bed and sides at suitable moisture; build new embankments in about 15 cm layers with a top width of about 40–45 cm or more; allow a freeboard of at least 5 cm.
2. Keep side slopes generally no steeper than 1 : 1 (¾ : 1 for hand-placed concrete deeper than 60 cm). Vertical sides need forms; unreinforced vertical sides are unsuitable deeper than 45 cm.
3. Lay 4–5 cm thick lining in 2–3 m lengths with joints every 2–3 m to control shrinkage cracking. Cut contraction joints as a vertical groove in the top one-third — e.g. a 1.5 cm groove in a 4.5 cm lining — with a 4–6 mm flat steel piece, and fill them with a sealing compound such as bitumen.
4. Cure it by frequent sprinkling or wet gunny bags for full strength and durability and to prevent shrinkage cracks.

Precast and low-cost sections

- Precast units made centrally and carried to site suit low discharges (low cost, easy operation); commercial concrete half-pipes also serve. Laying is rapid — sections of 1 m or more are placed and the joints cemented — but freight and breakage in long hauls are drawbacks.
- **Low-cost U-sections** are cast in U-shaped wooden forms (Fig. 7.15) of 2.5 cm hardwood boards made by a local carpenter, with a 1 cement : 3 sand : 4 gravel mix and minimum water.
- **Casting (Figs 7.16–7.17):** place the outer form on a sand bed, lay and tamp a 2.5 cm bottom layer, set the inner form, fill and tamp the sides, pull the pins and remove the forms; air-dry the sections 24–48 hours and cure them 1 week or more; lay them on an earthen embankment, seal the joints with cement mortar and coat the inside with cement–water to make it leak-proof.

Brick or stone masonry lining: single-layer brick or tile, rubble or flat stone slabs laid flat on the compacted bottom with 1 : 4 cement-mortar joints; plaster the inside if watertight joints are hard to get. Choose by availability and cost.

Designing a field channel

The channel must have good hydraulic properties, enough capacity for the design flow and a velocity that neither scours nor silts. Design factors: shape, depth of flow, bed slope, desirable velocity, and bottom width and side slopes (trapezoidal).

1. Fix the discharge to be carried.
2. Choose the bed slope from the topography.
3. Select a cross-section.
4. Compute the velocity (Manning) for the assumed section.

5. If it is too high or too low, change the section; alter the slope only as a last resort.

Sources: ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The theoretically most efficient channel section is

(a) rectangular (b) triangular
(c) semicircular (d) trapezoidal

Q2. Best rectangular channel has depth equal to

(a) breadth (b) ½ breadth
(c) ¼ breadth (d) 2 × breadth

Q3. Maximum permissible velocity in a sandy loam earth channel is about

(a) 45 cm/s (b) 60 cm/s
(c) 90 cm/s (d) 150 cm/s

Q4. Silting begins in earth channels when bed slope falls below about

(a) 0.05% (b) 0.5%
(c) 1% (d) 2%

Q5. Typical thickness of concrete lining for field channels is

(a) 1–2 cm (b) 4–5 cm
(c) 10–15 cm (d) 25 cm

Q6. Contraction-joint grooves are cut to a depth of about

(a) the full thickness (b) one-third of the thickness
(c) one-tenth (d) none

Answers

Q1 — (c) semicircular
Q2 — (b) ½ breadth
Q3 — (b) 60 cm/s
Q4 — (a) 0.05%
Q5 — (b) 4–5 cm
Q6 — (b) one-third of the thickness — *e.g. 1.5 cm in a 4.5 cm lining.*